

Pooja Kiran

Security Engineer | AI & Agent Security | Application Security | Cloud IAM Security | Model Supply-Chain Security
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EXPERIENCE

Independent AI Security Researcher & Engineer

Aug. 2024 - Present

Self-Directed Research

Tempe, AZ, USA

- Developed threat models for 3 self-directed security research projects, translating agent-tool, IAM, and model-artifact risks into enforceable controls and regression tests.
- Established reproducible Python regression testing and GitHub Actions checks across 3 repositories; dated 2026 evidence includes **752 gateway tests**, **25 IAM rule IDs**, and **260 model-scanner tests**.

Business & Compliance Lead, AEROSSEC

Aug. 2025 - Dec. 2025

Honeywell Aerospace Technologies × Arizona State University

Tempe, AZ, USA

- Led business/compliance analysis for a proposed airline PSS protection layer and built a **\$120K first-year commercialization scenario** with a five-year financial model presented to ASU and Honeywell stakeholders.

Graduate Teaching Assistant, IT Grader

Jan. 2025 - Oct. 2025

Ira A. Fulton Schools of Engineering, Arizona State University

Mesa, AZ

- Evaluated approximately **85 undergraduate web-development submissions per semester** and **52 graduate cybersecurity assignments**, giving evidence-based feedback on security controls, policy, compliance, risk analysis, and implementation quality.

PROJECTS

Project periods are candidate-reported; metrics cite dated 2026 CI snapshots.

MCP Agent Security Gateway | *Python, FastAPI, MCP/JSON-RPC 2.0, Elastic Security*

Oct. 2025 - Sep. 2026

- Engineered a default-deny MCP/JSON-RPC 2.0 gateway enforcing capability authorization, PII/exfiltration checks, anti-SSRF policy, and rate limits before downstream tool execution.
- Implemented **69 current prompt-injection patterns** and tamper-evident audit logging to capture reviewable tool-call decisions and security telemetry.
- Latest Docker CI: **752 passing tests**; earlier snapshot: **81.91% statement coverage**. Includes **9 Elastic rules** and **21 core SIEM tests**.

AWS Agent Identity Guard | *Python, AWS IAM, SARIF 2.1.0*

Apr. 2025 - Sep. 2025

- Built a static IAM analyzer with **25 deterministic rules** for wildcard grants, `iam:PassRole`, `sts:AssumeRole`, privilege escalation, trust-policy weaknesses, audit tampering, and permission boundaries.
- Integrated 3 finding formats (text, JSON, SARIF 2.1.0) with GitHub Code Scanning to make policy risks visible in code review.
- Verified 2026 CI snapshot: **247 collected**, **244 passed**, **3 skipped**; the skipped checks require live AWS credentials.

HF Model Provenance Scanner | *Python, SafeTensors, GGUF, ONNX, Keras*

Nov. 2024 - Mar. 2025

- Developed a non-executing model supply-chain scanner for unsafe serialization and provenance indicators, avoiding untrusted code execution in its default inspection path.
- Combined pickle-opcode, AST/taint, dependency, and obfuscation checks across 5 model format families: SafeTensors, GGUF, ONNX, Keras, and pickle-derived artifacts.
- Verified 2026 CI snapshot: **260 passed**, **1 skipped at 75.67% statement coverage**; detected **33/33 committed adversarial fixtures**.

EDUCATION

Arizona State University

Tempe, AZ, USA

M.S., Information Technology (Security), GPA: 3.87/4.00

Aug. 2024 - May 2026

- Academic Participation:** Student participant, ASU, 2025; Generative AI learning assistant for personalized education.

M. S. Ramaiah University of Applied Sciences

Bengaluru, KA, India

B.Tech., Computer Science & Engineering, CGPA: 8.44/10

Aug. 2019 - Aug. 2023

TECHNICAL SKILLS

AI & Agent Security: MCP/JSON-RPC 2.0, Tool/Function-Call Security, Prompt Injection, LLM Red Teaming, Model Supply-Chain Security

Identity & Application Security: Cloud IAM Security, AWS IAM, Least Privilege, Trust Policies, Permission Boundaries, Capability Authorization, Threat Modeling, API Security

Security Engineering & DevSecOps: Python, FastAPI, pytest, GitHub Actions, CI/CD Security, SARIF 2.1.0, GitHub Code Scanning, Docker

Detection & Observability: Elastic Security, ECS, SIEM, Prometheus, Tamper-Evident Audit Logging, MITRE ATLAS

CERTIFICATIONS

AWS Academy Graduate - Cloud Architecting (Apr. 2025) | AWS Academy Graduate - Cloud Security Foundations (Nov. 2025)